Methodology Report

Visualisation and Analysis on Namma Yatri Data

Exploratory data analysis and data storytelling in Tableau Public

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Executive summary

Namma Yatri recorded 2,161 ride searches in Bengaluru, of which 983 became completed rides (45.5%) and earned Rs 7,51,343 (average fare Rs 764). The analysis shows that the business has a conversion problem, not a timing problem.

- **Demand and revenue are flat across the day.** Completed trips average 41 an hour, and every time period earns 15.3% to 18.1% of revenue. Hour of day has no relationship with fare (r = -0.01), so peak-hour or surge tactics have no support in the data.
- **Most losses happen before a ride starts.** 40.9% of searches never get a driver. Of the 1,277 rides that did get one, 77.0% finished, 12.3% were cancelled by the customer and 10.7% by the driver. Overall, 67.6% of searches that asked for quotes ended in a completed trip.
- **Zones differ far more than hours.** Zone conversion ranges from 31.7% (Anekal) to 58.5% (Chamrajpet). Mahadevapura has the most requests (75) but only 26 completed trips, while Bangalore South earns the most revenue (Rs 30,295) from 57 requests. The top five zones by revenue bring only 18.7% of revenue, so growth has to come from many zones.
- **Zone leadership changes with the time of day.** The top five zones for the full day differ from the top five at night, so supply should be planned by zone and period, not by one city-wide rule.
- **Payments are already mostly digital** (76.1% of trips), and customers use all four methods almost equally.

Recommended actions: raise driver allotment from 59% to 70% with acceptance incentives and fast re-assignment; fix the eight weakest-converting zones first; position drivers by zone and time period; move the remaining cash users to UPI and cards with cashback; and target promotions at high-fare, low-volume zones. Detailed recommendations, KPIs and supporting dashboards are in Section 3.

Approach and assumptions

A. Problem statement

Namma Yatri runs an auto-rickshaw ride-hailing service in Bengaluru. Management wants to (1) find when and where demand is highest, (2) understand why searches do not turn into completed rides (cancellations and unallotted requests), (3) understand payment behaviour and trip economics, and (4) use these findings to allocate drivers better and sharpen marketing. This report analyses 2,161 trip searches (Trip_Details) of which 983 became completed trips (Trips), across 37 Bengaluru assembly zones.

B. Overall approach (storytelling structure)

Story element	What was done in this project
Context	Bengaluru auto-rickshaw demand; goal = more completed rides, higher revenue, fewer cancellations.
Data	5 tables joined into one master dataset; funnel columns (estimate, quote, allotment, cancellation, OTP, end ride) and trip columns (fare, distance, hour, payment, zone).
Visuals	One chart per question, chosen for the data type: line/bar for time, donut for share, scatter for correlation, bar for ranking, heatmap for zone x time, funnel for conversion.
Narrative	Demand and revenue are evenly spread across the day, so time-of-day is not the lever. The real problems are the booking funnel (only 45.5% of searches complete) and large zone-to-zone differences in conversion (32% to 58%).
Action	Fix supply and conversion in specific zones, reduce post-allotment cancellations, and target marketing at high-fare, low-volume zones.

C. Tools and method

- Tool logic: Tableau (relationships between the five tables, calculated fields, parameter). Every calculated field is written out in Tableau syntax under the relevant task so it can be rebuilt one-to-one.
- Software: all charts in this report marked (Tableau) were built in Tableau Public from the cleaned master file and the five-table Excel source. Data checks, the cleaned master file and the charts marked Supplementary were prepared in Python (pandas, matplotlib) using the same definitions, and the totals agree with Tableau (2,161 searches, 983 completed trips, Rs 7,51,343 revenue).
- Steps: (1) import the five tables and relate them in Tableau (relationships), (2) validate and clean, (3) classify variables, (4) univariate, segmented-univariate and bivariate analysis, (5) funnel analysis, (6) parameter-driven view, (7) recommendations supported by dashboards.
- Key definitions/assumptions: Revenue = sum of fare on completed trips (fare exists only for completed rides). Demand = number of completed trips for time analysis (the only table with an hour) and number of searches for zone request analysis. Hour = Duration id - 1 (id 1 = 0-1 AM).
- Time periods used: Late Night 0-4, Early Morning 4-8, Morning 8-12, Afternoon 12-16, Evening 16-20, Night 20-24.

D. Assumptions

- Each Trip_Details row is one customer search (searches = 1 on every row); one tripid = one search.
- A row in Trips = a completed ride; fare is the final fare and revenue = sum of fares on completed rides (no refunds, taxes or driver commission information).
- Demand over time = completed trips per hour, because searches carry no hour. Zone request demand = searches by pickup zone (loc_from).

- Trips.duration is the hour-slot id (1 = 0-1 AM), so Hour = id - 1. All trips are assumed to belong to a single typical period (no dates in the data), so no weekday or seasonal effect is analysed.
- Cancellation flag value 0 counts as a cancellation only when a driver was allotted (searches_got_quotes = 1); otherwise it means "not applicable".
- Time periods are defined as Late Night 0-4, Early Morning 4-8, Morning 8-12, Afternoon 12-16, Evening 16-20, Night 20-24.
- Fare outliers and fares not related to distance are retained and flagged, not removed, since they cannot be verified.
- "Other Assemblies" is treated as one zone. Where two zones tie (Gandhi Nagar and Rajarajeshwarinagar, 32 trips) the tie is stated.
- Per-period conversion is shown only as an illustration under an equal-searches assumption (see 2.9.2).

E. How the analysis types map to the tasks (and why each variable matters)

Analysis type	Where used	Business meaning
Univariate (one variable)	2.2 trips by hour; 2.3 revenue by period; 2.5 payment methods; 2.6 zones by requests / revenue; fare and distance distributions	Describes the normal level and spread: average 41 trips an hour, Rs 764 fare, 14 km ride. Shows whether demand is concentrated or spread out.
Segmented univariate (one measure split by a category)	2.7 trips by zone and period; 2.9.1 cancellations by zone (hotspot chart); revenue by time period; trips by distance band; zone and customer segments (segment chart)	Shows where the average hides big differences: cancellations of 9% to 42% by zone, conversion of 32% to 58% by zone, customers who give 43% of revenue.
Bivariate (two variables together)	2.4 hour vs fare (r = -0.01); hourly trips vs revenue (r = 0.90); fare vs distance (r = 0.006); requests vs conversion by zone (zone conversion chart)	Tests what drives what: revenue is driven by trip count, not by hour or distance, so volume and conversion are the levers rather than price.

Why the variables matter: hour and time period tell operations when to staff; pickup zone says where; the cancellation and quote flags show where riders are lost; fare and distance show the trip economics; payment method shows customer behaviour; driver and customer IDs show how concentrated supply and demand are.

1. Data Preparation

1.1 Import and Join Tables Correctly

1.1.1 Import

The workbook with five sheets was connected as one Excel data source. Row counts: Assembly 37, Duration 24, Payment 4, Trip_Details 2,161, Trips 983.

1.1.2 Relationship design (Tableau: Data Source tab > drag tables onto the canvas)

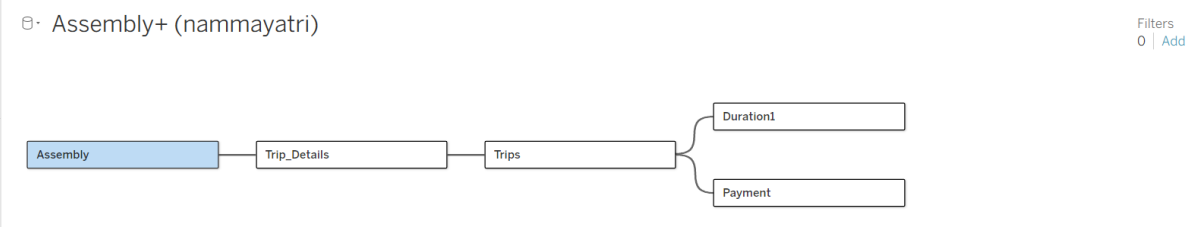


Figure 1: Tableau Data Source page: five tables related on their key fields (Assembly - Trip_Details - Trips - Duration1 / Payment).

Tableau relationships keep every table at its own level of detail and join only the tables a chart needs. The table below lists the key matches; the equivalent join types are shown for reference, and a drop-zone name copy of Assembly was added in the cleaned master file.

Table 1.1: Relationship (key) design

Join #	Left table	Right table	Type	Join clause	Result
1	Trip_Details	Trips	LEFT	Trip_Details.tripid = Trips.tripid	2,161 rows; 983 matched (completed rides), 1,178 have no trip record
2	Trip_Details	Assembly (pickup)	INNER	Trip_Details.loc_from = Assembly.ID	Pickup zone name; all 2,161 matched
3	Trips	Assembly (drop) - 2nd copy	LEFT	Trips.loc_to = Assembly (2).ID	Drop zone name for completed trips
4	Trips	Payment	LEFT	Trips.faremethod = Payment.id	Payment method name
5	Trips	Duration	LEFT	Trips.duration = Duration.id	Hour label (e.g. 13-14)

Trip_Details is the base (left) table because it holds every search; Trips only has completed rides. An inner join would have dropped 1,178 searches and made cancellation and conversion analysis impossible.

1.1.3 Relationship validation

- tripid is unique in both Trip_Details (2,161) and Trips (983); no duplicate keys, so the joins do not multiply rows (row count stays 2,161 after all joins).
- Every Trips.tripid exists in Trip_Details (0 orphans). Every loc_from, loc_to, faremethod and duration value exists in its lookup table.
- Trips.loc_from equals Trip_Details.loc_from for all 983 matched trips, so one pickup-zone field is enough.
- Control totals after joining: 2,161 searches; 983 completed; total fare Rs 7,51,343.

1.2 Find and Resolve Inconsistencies

1.2.1 Checks performed and result

Table 1.2: Inconsistencies found and how they were resolved

Check	Finding	Action
Missing values	None in any table before the join. After the left join, fare, distance, driver, customer, payment, hour are blank for the 1,178 non-completed searches.	Expected (no trip happened). Kept as NULL; revenue and hour analysis filtered to end_ride = 1.
Duplicates	0 duplicate rows, 0 duplicate tripid.	None needed.
Cancellation flags (key issue)	customer_not_cancelled / driver_not_cancelled = 0 appears on 1,041 / 1,021 rows, but 884 of them were never allotted a driver (flags are 0 by default). Reading 0 as "cancelled" would wildly overstate cancellations (48% instead of 12%).	Cancellation calculated only on rows where a driver was allotted (searches_got_quotes = 1, n = 1,277).
Funnel logic	No impossible sequences: no quote without estimate, no allotment without quote search, no OTP without allotment; otp_entered equals end_ride on all rows; no completed ride is flagged cancelled.	None needed.
Constant column	searches = 1 on every row (each row is one search).	Used as a row counter only.
Misleading column name	Trips.duration is the hour-slot id (1-24), not minutes.	Renamed Hour ID; Hour = id - 1.
Fare outliers	Fare ranges Rs 44-1,500; fare per km ranges Rs 1.9-739.5 (e.g. Rs 1,479 for 2 km). Correlation of fare with distance is only 0.006, which is not realistic for metered autos.	Not deleted: values cannot be verified and removing them would bias revenue. Flagged as a data-quality risk; analysis uses totals and medians where appropriate.
Pickup = drop	30 completed trips start and end in the same zone.	Kept (valid short local rides).
Category labels	Payment names in lower case; zone "Other Assemblies" is a catch-all (31).	Payment names title-cased; "Other Assemblies" kept as its own zone and noted as non-specific.
Lookup text mismatch	Starter file names Assembly_ID / duration_id; the file has ID / id.	Mapped by meaning.

1.2.2 Calculated fields (Tableau syntax)

```
Hour          = INT(LEFT([Duration], FIND([Duration],"-")-1))
  (or [Trips.duration] - 1 when joining on id)
Time Period   = IF [Hour]<4 THEN "1 Late Night (0-4)" ELSEIF [Hour]<8 THEN "2 Early Morning (4-8)"
              ELSEIF [Hour]<12 THEN "3 Morning (8-12)" ELSEIF [Hour]<16 THEN "4 Afternoon (12-16)"
              ELSEIF [Hour]<20 THEN "5 Evening (16-20)" ELSEIF [Hour]>=20 THEN "6 Night (20-24)" END
  (blank Hour stays NULL: searches that never became a trip have no hour)
Completed     = IF [end_ride]=1 THEN 1 ELSE 0 END
Revenue       = SUM(IF [end_ride]=1 THEN [fare] END)
Ride Outcome  = IF [end_ride]=1 THEN "Completed"
              ELSEIF [searches_got_quotes]=1 AND [customer_not_cancelled]=0 THEN "Customer cancelled"
              ELSEIF [searches_got_quotes]=1 AND [driver_not_cancelled]=0 THEN "Driver cancelled"
              ELSE "Not allotted" END
Fare per km   = [fare] / [distance]
```

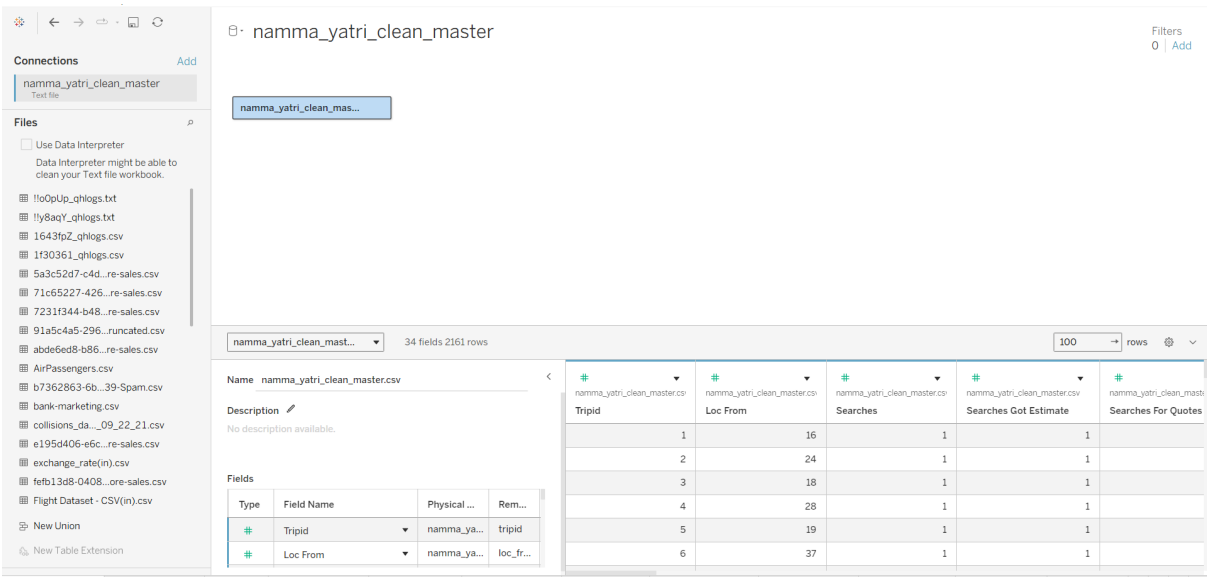


Figure 2: Tableau: the cleaned master file namma_yatri_clean_master.csv (34 fields, 2,161 rows) used for the analysis.

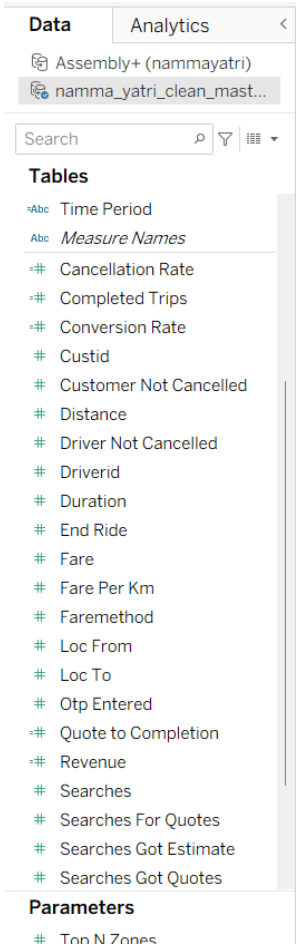


Figure 3: Tableau Data pane showing the calculated fields created (Time Period, Revenue, Completed Trips, Ride Outcome, Conversion Rate, Quote to Completion, Cancellation Rate).

Outcome: a validated, single master table of 2,161 rows (also supplied as namma_yatri_clean_master.csv) that every later chart uses. No rows were deleted.

2. Exploratory Data Analysis

2.1 Classify Variables into Categorical and Numerical

Table 2.1: Variable classification

Variable	Table	Type	Sub-type / role
tripid	Both trip tables	Categorical	Identifier (not analysed)
Assembly_ID / loc_from / loc_to / Assembly (zone name)	Assembly, Trips, Trip_Details	Categorical	Nominal - pickup and drop zone
duration (hour id) / hour label / time period	Duration, Trips	Categorical	Ordinal - hour slot 0-1 ... 23-24; derived period
method / faremethod	Payment, Trips	Categorical	Nominal - Cash, UPI, Debit, Credit
driverid, custid	Trips	Categorical	Identifiers (30 drivers, 99 customers)
searches_got_estimate, searches_for_quotes, searches_got_quotes, customer_not_cancelled, driver_not_cancelled, otp_entered, end_ride	Trip_Details	Categorical	Binary (0/1) funnel flags
searches	Trip_Details	Numerical	Discrete count (always 1)
fare	Trips	Numerical	Continuous (Rs 44-1,500, mean 764)
distance	Trips	Numerical	Continuous (2-27 km, mean 14.4)
Derived: Revenue, Completed trips, Conversion %, Fare per km	Calculated	Numerical	Measures

Why it matters: the flags are categorical in type but behave as numbers when summed (SUM of end_ride = completed trips). Treating them correctly is what lets us build conversion rates. IDs are never averaged or plotted as measures. In Tableau, dimensions are blue pills (zone, period, payment) and measures are green pills (fare, distance, trips).

2.2 Analyse Ride Demand Over Time

Demand parameter chosen: completed trips per hour (count of tripid in Trips). It is the only demand measure that carries a time stamp; searches have no hour.

```
Tableau: Columns = [Hour] (converted to Dimension), Rows = SUM([End Ride]) (= completed trips); Mark = Line.
Second sheet: Columns = [Time Period], Rows = COUNTD([tripid]); Mark = Bar.
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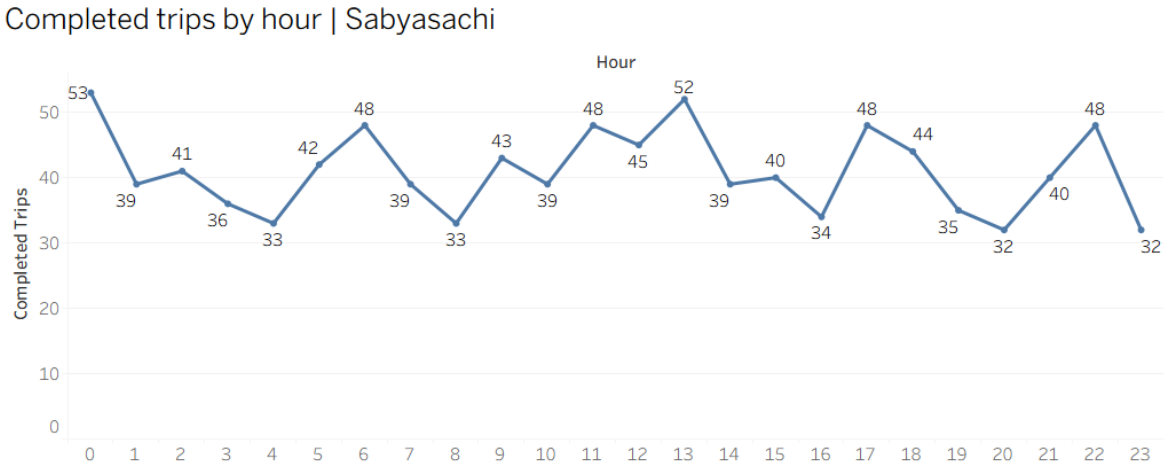


Figure 4: Completed trips by hour (Tableau).

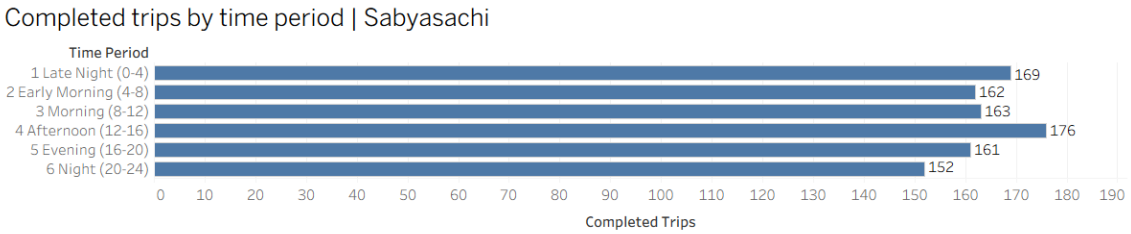


Figure 5: Completed trips by time period (Tableau).

Insights

- Demand is flat: average 41 trips per hour and no hour is more than ~30% above or below it. There is no classic morning or evening rush.
- Peak hours are 0:00-1:00 (53 trips) and 13:00-14:00 (52 trips); 6:00, 11:00, 17:00 and 22:00 follow with 48 each. The quietest hours are 20:00 and 23:00 (32) and 4:00 and 8:00 (33).
- By period, Afternoon (12-16) is busiest with 176 trips and Night (20-24) the quietest with 152, a gap of only 16%.
- **Business meaning:** driver supply cannot be scheduled around one or two rush windows. A round-the-clock roster with small boosts at midnight and 1 PM is enough; bigger gains lie in zones and conversion (see 2.6-2.9).

2.3 Proportion of Total Revenue from Different Time Periods

Tableau: Rows = [Time Period]; Columns = Revenue shown as % of Total (axis); Label = Revenue in Rs; Mark = Bar.
Calculated field: Revenue = SUM(IF [End Ride]=1 THEN [Fare] END)

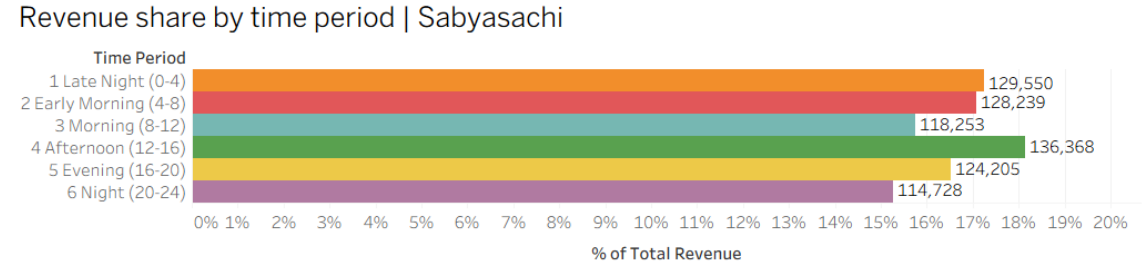


Figure 6: Revenue share by time period (bar length = % of total revenue; labels = Rs; total Rs 7,51,343).

Table 2.3: Revenue by time period

Time period	Trips	Revenue (Rs)	Share of revenue	Avg fare (Rs)
Late Night (0-4)	169	1,29,550	17.2%	767
Early Morning (4-8)	162	1,28,239	17.1%	792
Morning (8-12)	163	1,18,253	15.7%	725
Afternoon (12-16)	176	1,36,368	18.1%	775
Evening (16-20)	161	1,24,205	16.5%	771
Night (20-24)	152	1,14,728	15.3%	755

Insights

- Afternoon contributes the most (18.1%), Late Night (17.2%) and Early Morning (17.1%) follow. Morning (15.7%) and Night (15.3%) contribute the least.
- The spread between best and worst period is only 2.8 percentage points: revenue is not concentrated in a peak, so no period can be ignored.
- Early Morning has the highest average fare (Rs 792) despite fewer trips; Morning has the lowest (Rs 725). Morning is a candidate for fare and incentive review.

2.4 Explore the Relationship Between Trip Hour and Revenue

Tableau: Columns = [Hour], Rows = SUM([fare]); Mark = Bar (top 4 coloured).
Correlation: Analytics pane > Trend Line (linear) on a scatter of [Hour] vs [fare] (Describe Trend Model shows R-squared),
or calculated field: CORR([Hour],[fare]).

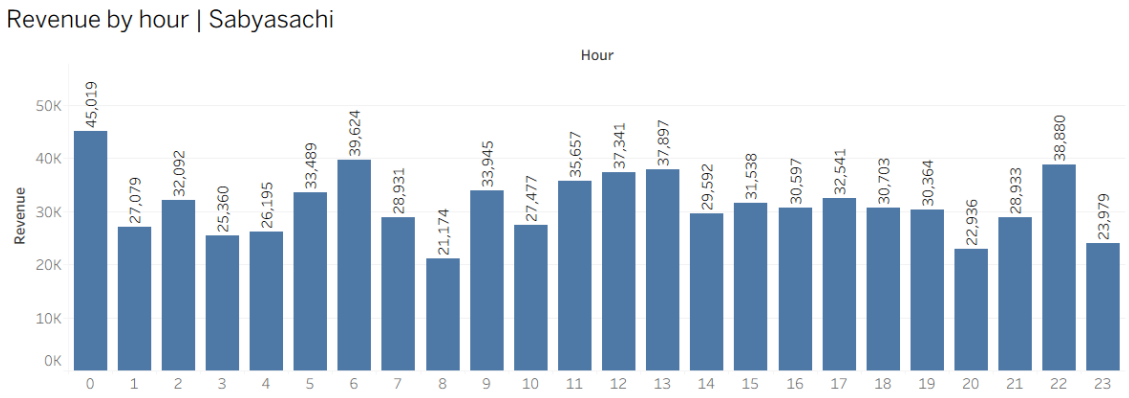


Figure 7: Revenue by hour (Tableau).

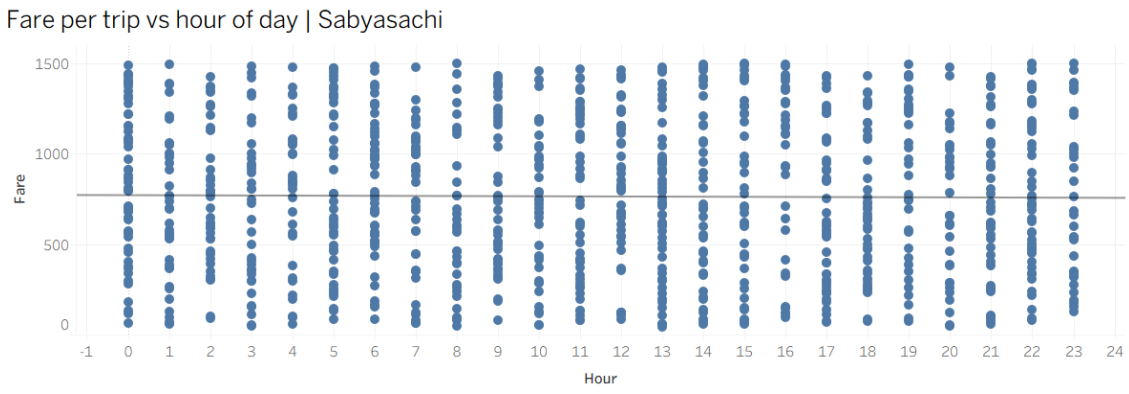


Figure 8: Fare per trip vs hour with linear trend line (Tableau): the line is flat.

Insights

- Correlation between trip hour and fare per trip is -0.01, i.e. none. Going later in the day does not make a ride more expensive.
- Hourly revenue is driven by how many trips happen, not by fare size: correlation of hourly trip count with hourly revenue is 0.90, versus 0.12 between trip count and average hourly fare.
- Highest-revenue hours: 0:00 (Rs 45,019), 6:00 (Rs 39,624), 22:00 (Rs 38,880), 13:00 (Rs 37,897). Lowest: 8:00 (Rs 21,174).
- **Business meaning:** to raise revenue in a given hour, add completed trips (supply, conversion), not price. Surge pricing by hour has no support in this data.

2.5 Examine the Popularity of Different Payment Methods

Tableau: Rows = [method]; Columns = COUNTD([tripid]) and SUM([fare]); Label = % of Total. Colour top method.

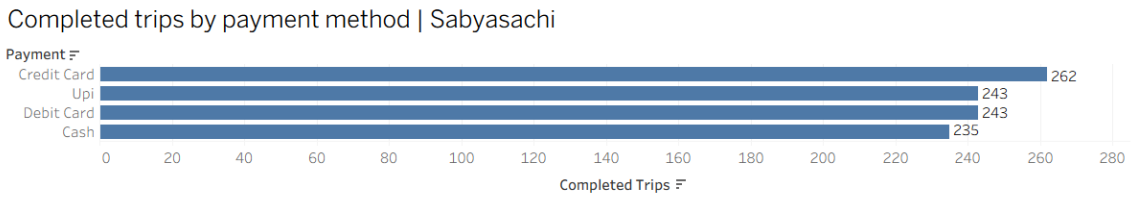


Figure 9: Completed trips by payment method (Tableau).

Table 2.5: Payment methods

Method	Trips	% of trips	Revenue (Rs)	Avg fare (Rs)
Credit Card	262	26.7%	1,97,941	756
Debit Card	243	24.7%	1,84,676	760
UPI	243	24.7%	1,83,624	756
Cash	235	23.9%	1,85,102	788

Insights

- Credit card is the most used method (26.7% of trips and 26.3% of revenue), but all four methods sit within 3 points of each other.
- 76.1% of rides are paid digitally, so cash is no longer dominant.
- Cash rides carry the highest average fare (Rs 788), only slightly above the rest.
- Ride frequency does not depend on payment method; customers use all methods. Business meaning: keep all four enabled; use UPI/card cashback (instead of cash discounts) to push the remaining 24% cash share digital and reduce driver cash handling.

2.6 Identify High-Performing Zones

2.6.1 Rides: Identify pickup zones with the highest number of trip requests.

Tableau: Rows = [Pickup Zone (Assembly)], Columns = SUM([searches]) (trip requests); sort descending; top 10 kept with Keep Only.

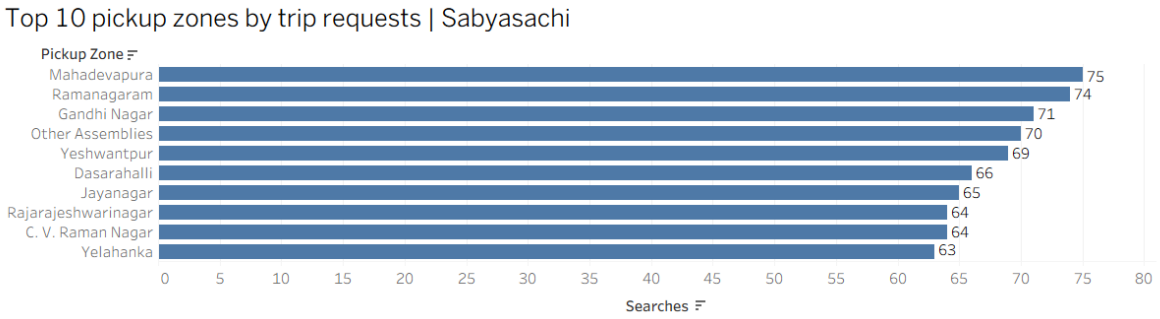


Figure 10: Top 10 pickup zones by trip requests (Tableau).

Insights

- Most requests: Mahadevapura (75), Ramanagaram (74), Gandhi Nagar (71), Other Assemblies (70), Yeshwantpur (69).
- Request volume is a poor predictor of completed rides. Mahadevapura has the most requests but only 26 completed trips (34.7% conversion): drivers are not available there or are unwilling to accept, so demand is being lost.
- Zones with 60+ requests are the first place to put extra drivers.

2.6.2 Revenue: Identify pickup zones generating the highest revenue.

Tableau: Rows = [Pickup Zone], Columns = Revenue (= SUM(IF [end_ride]=1 THEN [fare] END)); sort descending; Top 10.

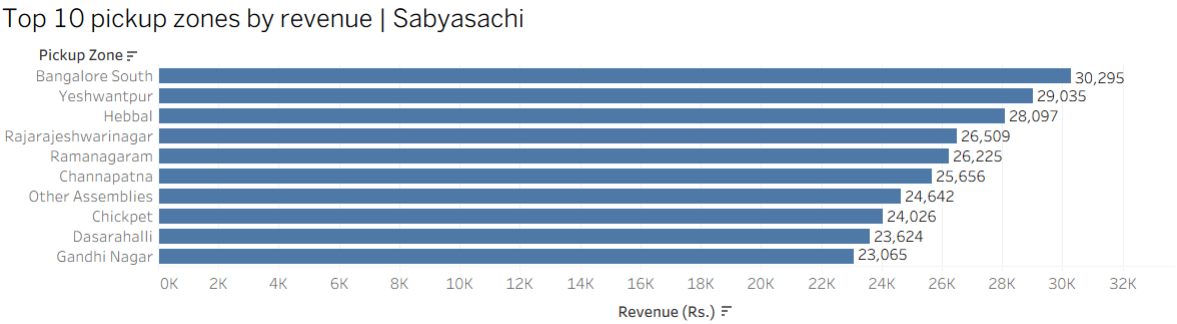


Figure 11: Top 10 pickup zones by revenue (Tableau).

Insights

- Top revenue zones: Bangalore South (Rs 30,295), Yeshwantpur (29,035), Hebbal (28,097), Rajarajeshwarinagar (26,509), Ramanagaram (26,225).
- Bangalore South earns the most with only 57 requests because it converts 57.9% of requests and has the highest average fare among large zones (Rs 918). Quality of conversion beats raw request volume.
- Mahadevapura, first in requests, is only 11th-plus in revenue (Rs 20,222): the clearest missed-revenue zone.

2.7 Analyse Ride Time Periods Across Zones

Tableau: Rows = [Pickup Zone], Columns = [Time Period], Mark = Square/Heatmap, Colour = COUNTD([tripid]) (end_ride=1).

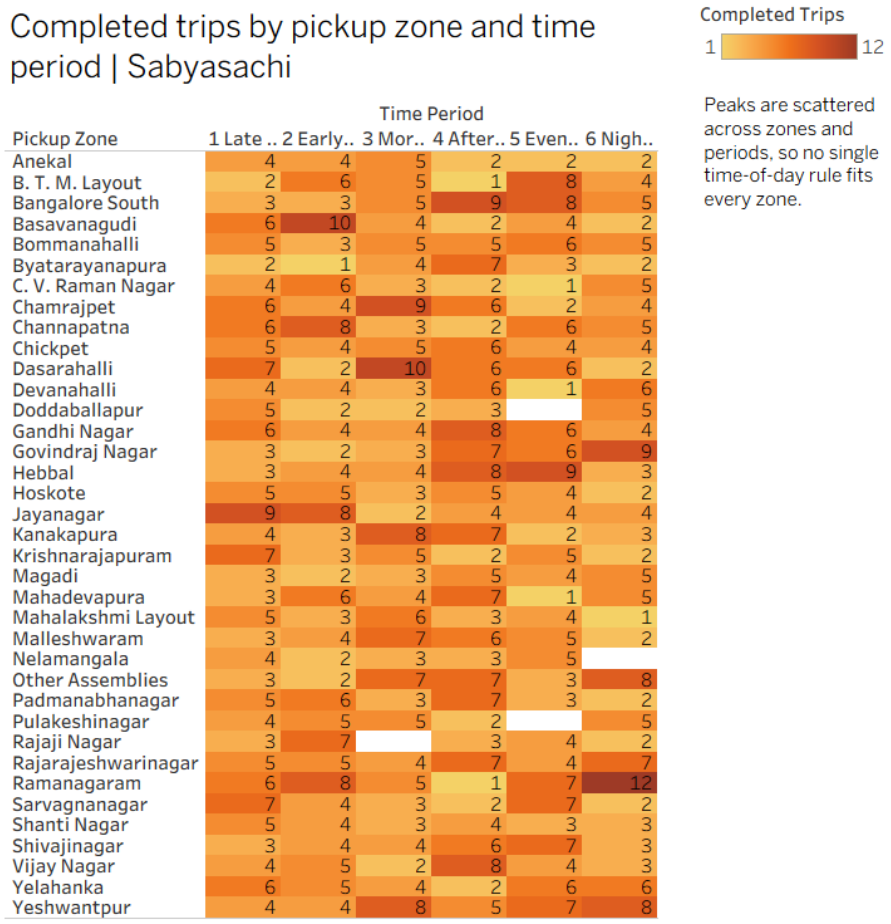


Figure 12: Completed trips by pickup zone and time period, all 37 zones (Tableau heatmap).

Insights

- Each zone has its own busy window, even though the city total is flat: Ramanagaram peaks at Night (12 trips) but has just 1 in the Afternoon; Dasarahalli peaks in the Morning (10); Jayanagar in Late Night and Early Morning (9 and 8); Bangalore South in the Afternoon (9); Hebbal in the Evening (9); Govindraj Nagar at Night (9); Chamrajpet in the Morning (9).
- Flat city-wide demand therefore hides zone-level swings: a city-wide shift roster would over-supply some zones and under-supply others.
- **Business meaning:** position drivers by zone-and-period (e.g., send drivers to Ramanagaram after 20:00 and to Hebbal at 16:00-20:00) rather than by hour alone.

2.8 Top Zones with Highest Trip Volume

Tableau: Filter [end_ride]=1; Rows = [Pickup Zone]; Columns = SUM([Completed]); Top 5 filter by SUM([Completed]).
Factors: add Conversion = SUM([end_ride])/COUNT([tripid]), AVG([fare]), AVG([distance]).

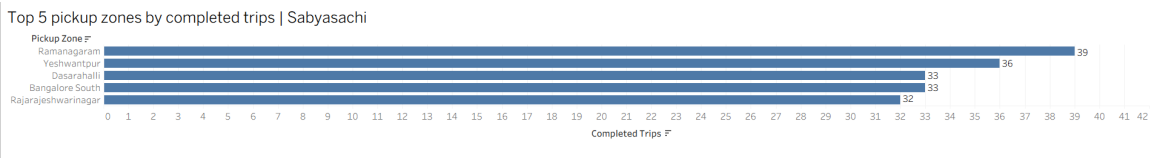


Figure 13: Top 5 pickup zones by completed trips (Tableau).

Table 2.8: Top-5 zones (Rajarajeshwarinagar also has 32 completed trips and ties with Gandhi Nagar for fifth place)

Zone	Requests	Completed	Conversion	Avg fare (Rs)	Avg dist (km)
Ramanagaram	74	39	52.7%	672	14.5
Yeshwantpur	69	36	52.2%	807	13.8
Dasarahalli	66	33	50.0%	716	13.6
Bangalore South	57	33	57.9%	918	13.7
Gandhi Nagar	71	32	45.1%	721	16.2

Insights

- Three factors explain the leaders: (1) high request volume (66-74 in four of the five), (2) conversion of 50-58% against the city average of 45.5%, and (3) typical trip length (about 14 km, no unusual long trips).
- Bangalore South is the efficiency star: 33 completed trips from only 57 requests (57.9%).
- Average fare and distance differ little between zones, so growth comes from volume and conversion, not from longer or pricier rides.
- Note on the tie: Gandhi Nagar and Rajarajeshwarinagar both have 32 completed trips and share 5th place. The Tableau chart shows Rajarajeshwarinagar (64 requests, 50.0% conversion); the table lists Gandhi Nagar (71 requests, 45.1% conversion). Both support the same conclusion.

2.9 Basic Analytical Tasks

2.9.1 What are the percentages of cancellations and successful rides by both driver and customer?

Ride Outcome (calculated field, see 1.2.2); Tableau: Rows = [Ride Outcome], Columns = COUNTD([tripid]) / TOTAL(COUNTD([tripid])).
For the driver-allotted base add a filter [searches_got_quotes] = 1.

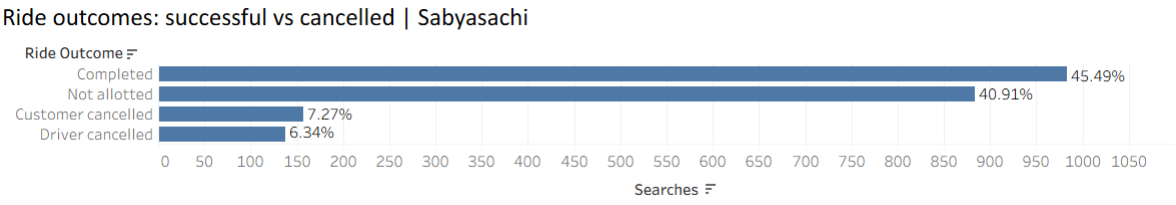


Figure 14: Ride outcomes as a % of all 2,161 searches (Tableau).

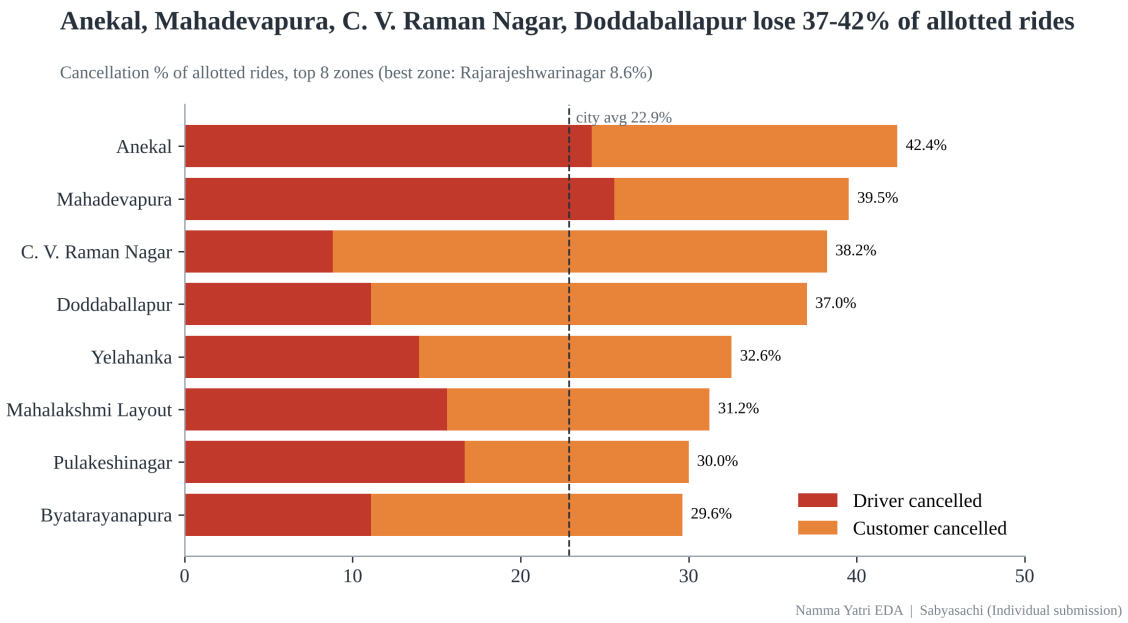


Figure 15: Supplementary (Python): cancellation hotspots: share of allotted rides cancelled, by pickup zone.

Cancellation hotspots (additional segmented analysis): zone cancellation ranges from 8.6% (Rajarajeshwarinagar) to 42.4% (Anekal) against a city average of 22.9%. Anekal (42.4%), Mahadevapura (39.5%), C. V. Raman Nagar (38.2%) and Doddaballapur (37.0%) are the hotspots. In Anekal and Mahadevapura most cancellations come from drivers (24% and 26% of allotted rides), which points to unattractive pickups or too few drivers. In C. V. Raman Nagar and Doddaballapur they come mainly from customers (29% and 26%), which points to long waits.

Table 2.9.1: Cancellations and successful rides

Outcome	Count	% of allotted rides (1,277)	% of all searches (2,161)
Successful (completed)	983	77.0%	45.5%
Cancelled by customer	157	12.3%	7.3%
Cancelled by driver	137	10.7%	6.3%
Not allotted a driver	884	-	40.9%

Insights

- Of rides where a driver was allotted, 77.0% succeed and 23.0% are cancelled: 12.3% by customers and 10.7% by drivers. Roughly one allotted ride in four is wasted.
- The bigger loss is before allotment: 40.9% of all searches never get a driver.
- Customers cancel slightly more than drivers, which usually points to long waits or a better offer elsewhere; the 10.7% driver cancellations suggest unattractive pickups (distance or fare).

2.9.2 Analyse the percentage of people who completed trips after searching for quotes. Visualise the variation of this ratio by time periods.

```
Quote-to-completion = SUM([end_ride]) / SUM([searches_for_quotes])
Allotted-to-completion = SUM([end_ride]) / SUM([searches_got_quotes])
Tableau: Rows = [Pickup Zone] / [Time Period]; Columns = the ratio above.
```

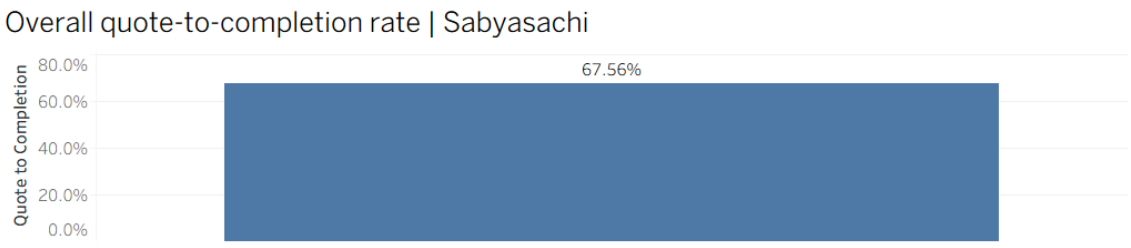


Figure 16: Overall quote-to-completion rate: 67.56% (Tableau).

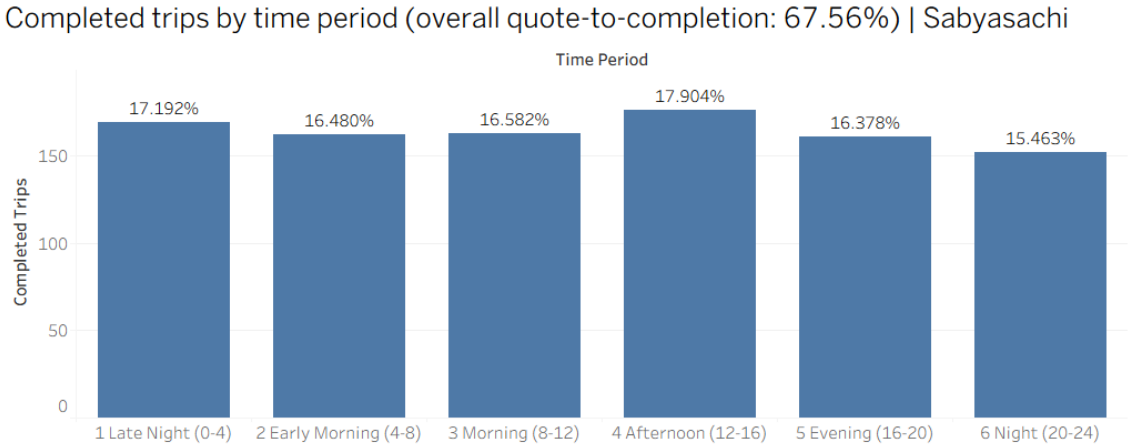


Figure 17: Completed trips by time period, with the overall rate in the title (Tableau).

67.6% of people who searched for quotes completed a trip; zone gaps are wide

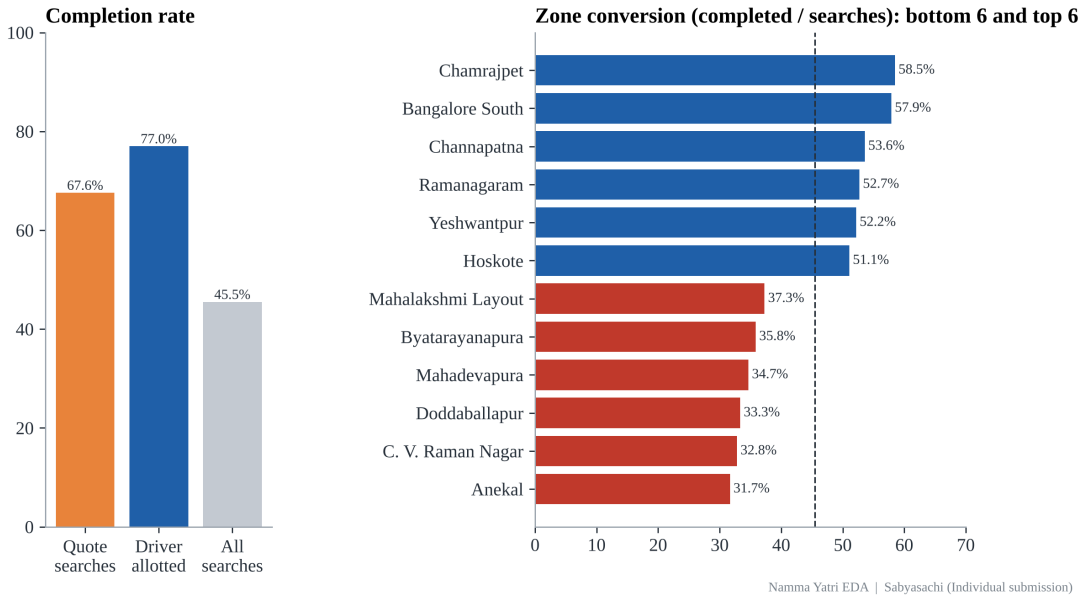


Figure 18: Supplementary (Python): share completing a trip after searching for quotes, plus zone variation.

Result

- Of 1,455 people who searched for quotes, 983 completed a trip: **67.6%**. Of 1,277 who were allotted a driver, 77.0% completed. Of all 2,161 searches, 45.5% completed.
- Funnel steps: estimate shown 81.4% -> searched for quotes 67.3% -> driver allotted 59.1% -> completed 45.5%. The biggest single drop is search to estimate (18.6% lost), then estimate to quote search (14 points).

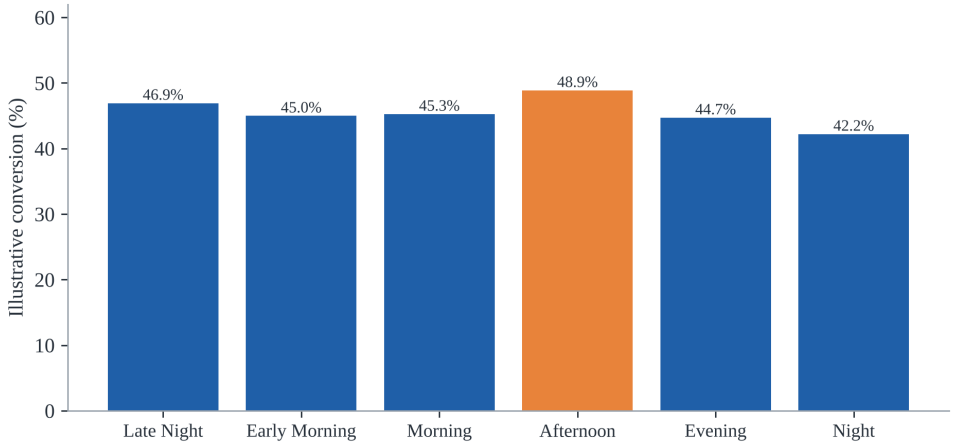
- By zone the ratio varies widely: lowest Anekal (31.7%), C. V. Raman Nagar (32.8%), Doddaballapur (33.3%), Mahadevapura (34.7%); highest Chamrajpet (58.5%), Bangalore South (57.9%), Channapatna (53.6%).

Variation by time period: data limitation and assumption

Only completed trips carry an hour field; searches that did not complete have no timestamp, so the exact completed/search ratio per period cannot be computed from this dataset. This is stated openly rather than inventing a number. As an illustration only, if searches were equal in every period (about 360 per period) the conversion would be as below. Completed trips by period (152 to 176) are the real, measured part.

Illustrative: conversion by period if searches were equal across periods

ASSUMPTION: Trip_Details has no timestamp, so each period is assumed to receive 1/6 of the 2,161 searches



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Figure 19: Illustrative only (Python): conversion by period under the equal-searches assumption, not measured.

- Under that assumption Afternoon is best (48.9%) and Night worst (42.2%).
- Recommendation: log a request timestamp in Trip_Details so this ratio can be measured exactly.

2.10 Create a Parameter and Use Filters

Parameter and filter created in Tableau

Control	Type	Allowed values	Default
Top N Zones	Parameter (Integer)	Range 3 to 15, step 1	5
Time Period	Quick filter on [Time Period]	Six periods (Null unticked)	All six ticked

Show Zone = RANK([Revenue]) <= [Top N Zones] (on the Filters shelf = True)
Rows = [Pickup Zone], Columns = Revenue (sorted descending); Time Period filter shown as a checklist; Top N Zones parameter shown as a slider.
Because the rank is a table calculation it is recomputed after the Time Period filter, so the Top N is always the top N of the selected periods.

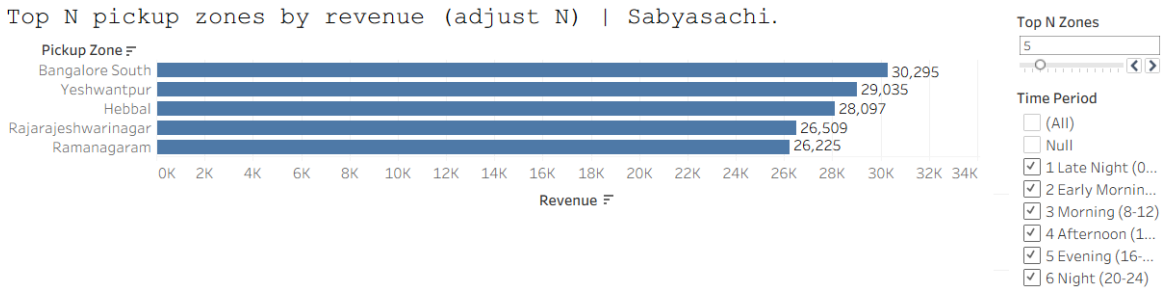


Figure 20: Top N = 5, all six periods selected (Tableau): same five zones as the all-day ranking.

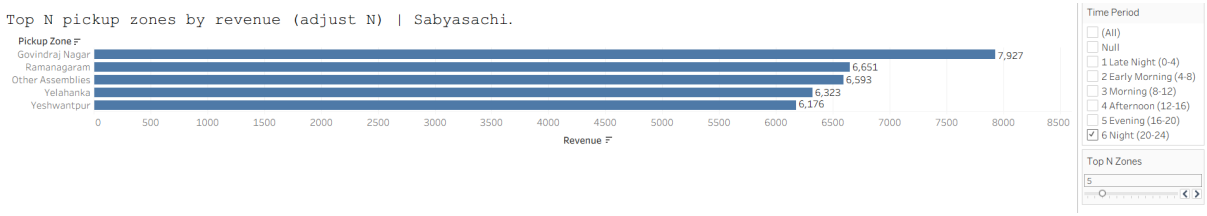


Figure 21: Top N = 5, only Night (20-24) selected (Tableau): the ranking changes.

Why this parameter and filter, and what they show

- Choice: zones (where) and time period (when) are the two levers operations control. The slider lets a manager look at the top 3 or the top 15 zones, and the period filter answers "which zones matter at this time of day?" without building a new chart each time. Revenue is the ranking measure because it combines volume, conversion and fare.
- All periods, N = 5: Bangalore South (Rs 30,295), Yeshwantpur (29,035), Hebbal (28,097), Rajarajeshwarinagar (26,509), Ramanagaram (26,225) = Rs 1,40,161, 18.7% of total revenue.
- Night only (20-24), N = 5: Govindraj Nagar (Rs 7,927), Ramanagaram (6,651), Other Assemblies (6,593), Yelahanka (6,323), Yeshwantpur (6,176) = Rs 33,670, 29.3% of night revenue (Rs 1,14,728).
- The leaders change with the time of day: Bangalore South, Hebbal and Rajarajeshwarinagar drop out of the night top 5, while Govindraj Nagar, Other Assemblies and Yelahanka enter. Only Ramanagaram and Yeshwantpur stay. Zone priorities are time-specific, which supports staffing by zone and period (see the heatmap in 2.7).

3. Conclusion

3.1 Recommendations for Operational Efficiency

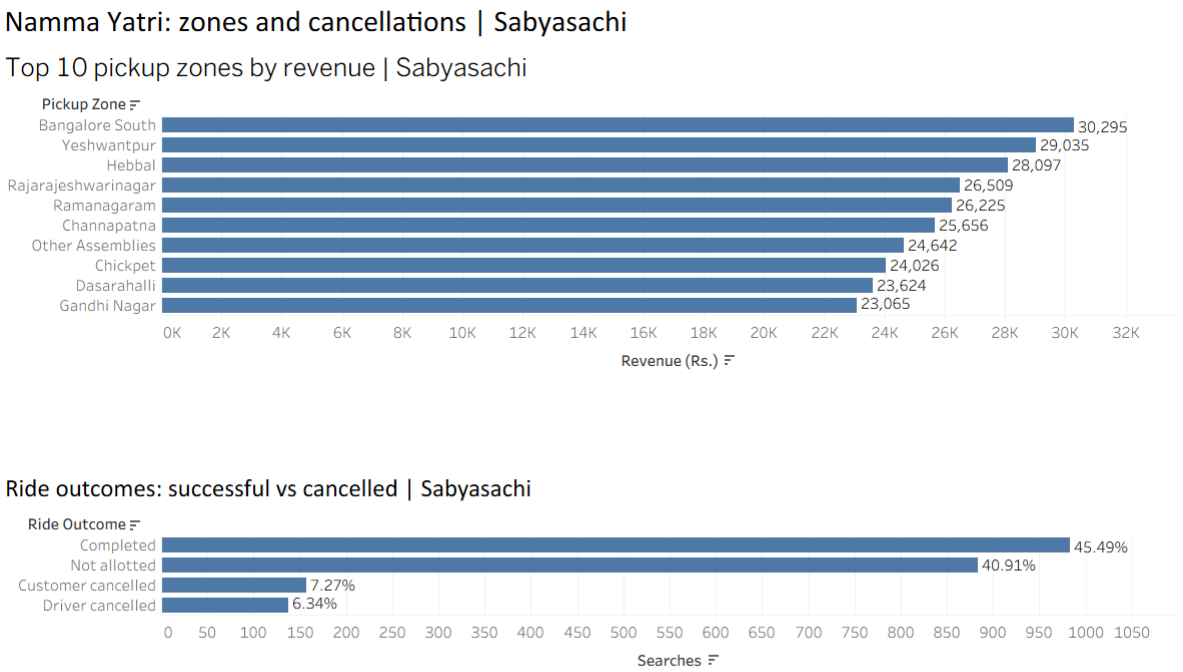


Figure 22: Tableau Dashboard 2 - zones and cancellations (supports the operations recommendations).

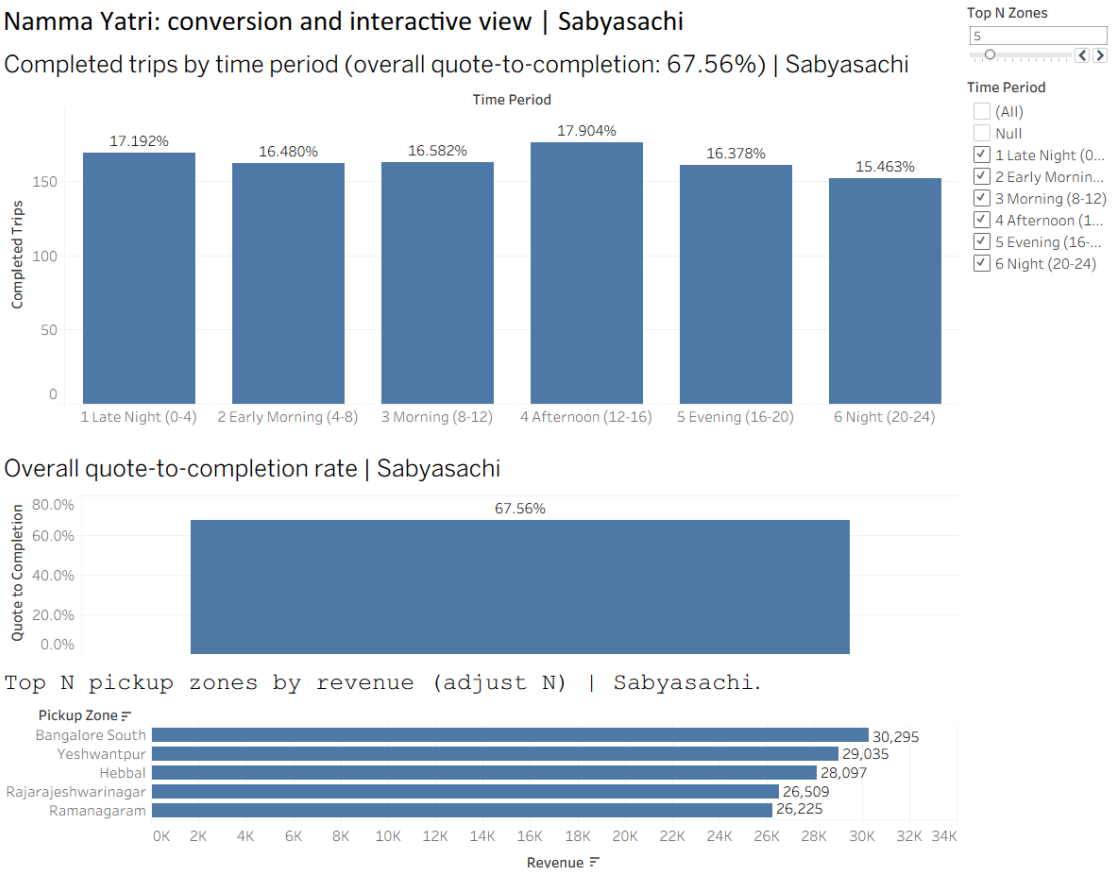


Figure 23: Tableau Dashboard 7 - conversion and interactive view (overall quote-to-completion 67.56%; Top N by period).

Table 3.1: Operational recommendations

#	Finding (evidence)	Recommendation	KPI to track
1	40.9% of searches are never allotted a driver; 1 in 4 allotted rides cancelled (12.3% customer, 10.7% driver).	Add driver-acceptance incentives (bonus for accepting within 30 s); show driver destination before accept; auto re-assign immediately after a cancellation.	Allotment rate 59% -> 70%; cancellation rate 23% -> 15%
2	Zone conversion ranges 31.7% to 58.5%. Mahadevapura (75 requests, 34.7%), Anekal, C. V. Raman Nagar, Doddaballapur are the weakest.	Create a "fix list" of 8 weakest zones; add dedicated drivers and pickup hotspots; review driver availability and accepted-pickup distance there.	Zone conversion >= 45% for all zones
3	City-wide demand is flat (152-176 trips per period) but each zone peaks in different periods (heatmap).	Dynamic zone-by-period driver positioning instead of one city roster; small boosts at 0:00 and 13:00.	Trips per driver-hour by zone
4	18.6% of searches never get an estimate and a further ~14% do not go on to search for quotes.	Investigate estimate failures (coverage, app errors, outside service area); simplify the estimate-to-quote step.	Estimate rate 81% -> 90%
5	Fare per km varies Rs 2-740 and is uncorrelated with distance: a 2-5 km ride averages Rs 716, a 21-27 km ride Rs 765.	Audit fare calculation; unreliable fares can drive customer cancellations and unattractive long rides for drivers.	Share of trips outside a fare-per-km band
6	Cancellation hotspots: Anekal 42%, Mahadevapura 40%, C. V. Raman Nagar 38%, Doddaballapur 37% (city 23%). Driver-led in the first two, customer-led in the last two.	Driver-led zones: pickup incentive and extra drivers. Customer-led zones: show accurate ETA, reduce wait, and re-match quickly.	Zone cancellation <= 20%

#	Finding (evidence)	Recommendation	KPI to track
7	Driver output is even (19-46 trips, average 33; top 10 drivers earn 39% of revenue). Driver 11 has only 19 trips.	No dependence risk on a few drivers. Reward the top tier (12, 24, 21) and coach the lowest (11, 26, 7). Set driver tiers on completion and cancellation.	Trips per driver, driver cancel rate

3.2 Marketing and Operational Strategy Improvements

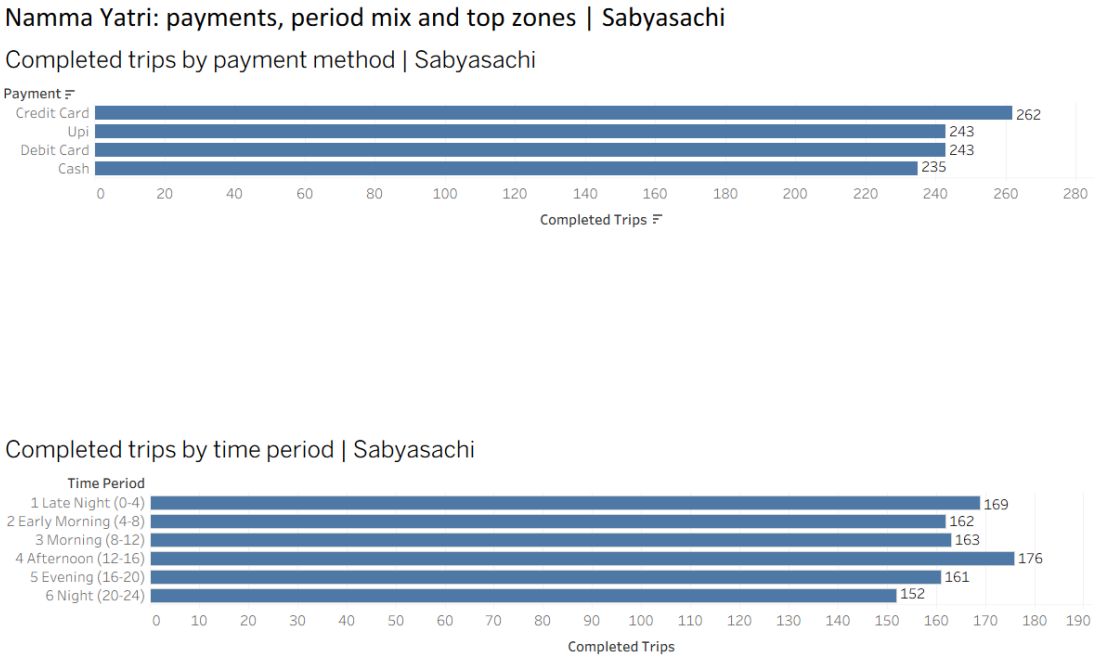


Figure 24: Tableau Dashboard 4 - payments and period mix (supports the marketing recommendations).

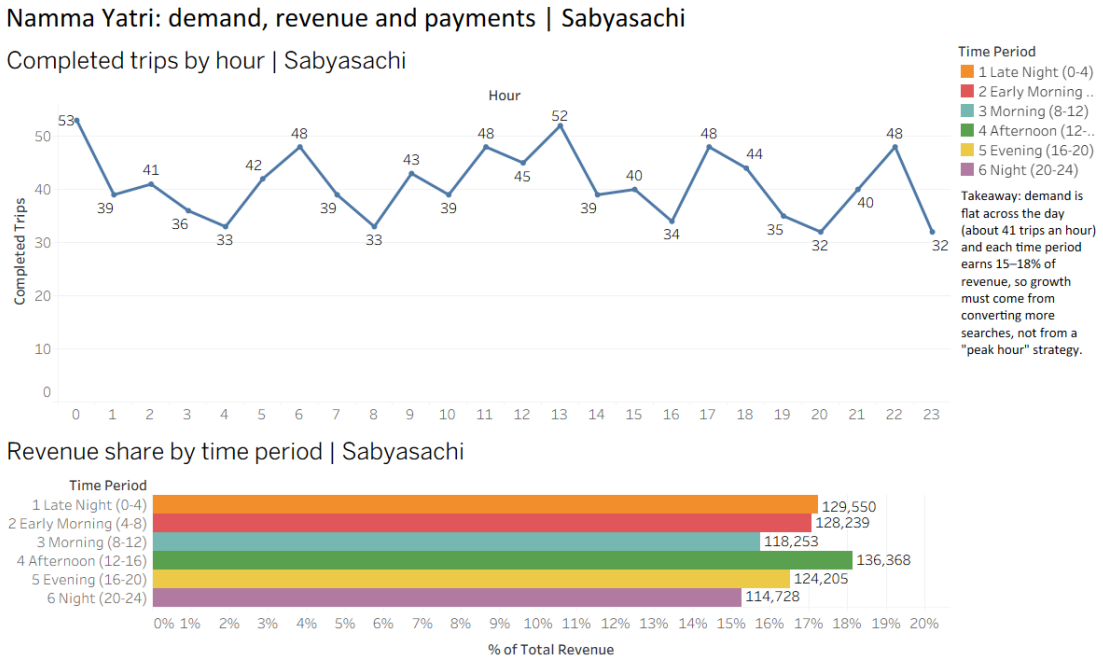


Figure 25: Tableau Dashboard 1 - demand by hour and revenue share by time period.

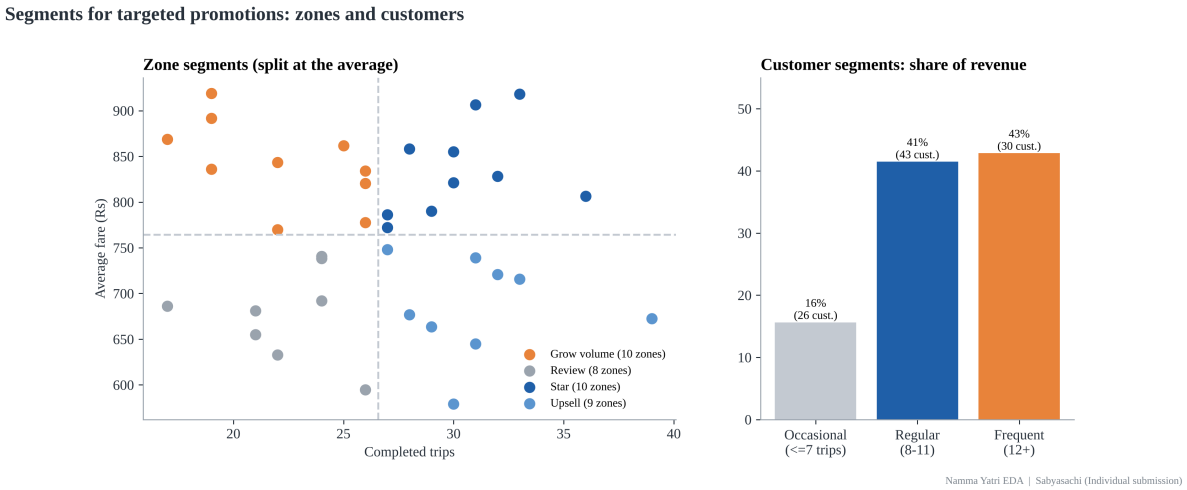


Figure 26: Supplementary (Python): segments for targeted promotions: zone segments (left) and customer segments (right).

Table 3.2: Marketing and strategy recommendations

#	Finding (evidence)	Recommendation	KPI to track
1	Revenue is spread evenly over the day (15.3%-18.1% per period); Morning and Night are weakest.	Off-peak promotions (e.g., Morning and Night ride-credit) instead of peak-hour surge; no evidence that price rises with hour.	Morning and Night revenue +10%
2	High-fare, low-volume zones: Byatarayanapura (Rs 919, 19 trips), Rajaji Nagar (892, 19), Doddaballapur (869, 17), Sarvagnanagar (862, 25), Anekal (836, 19).	Regional marketing and first-ride offers; driver incentives in these zones. A modest volume rise is worth more per trip.	Completed trips in these zones +25%
3	Bangalore South, Yeshwantpur, Hebbal are top revenue zones with high conversion.	Protect: keep driver supply high; loyalty offers for repeat customers.	Revenue per zone, repeat rate

#	Finding (evidence)	Recommendation	KPI to track
4	76% digital payments; credit card slightly leads; 24% still cash.	Cashback on UPI/cards to move cash users digital; cheaper, safer for drivers.	Digital share 76% -> 85%
5	Night top zones differ from all-day top zones (Top N parameter and period filter, 2.10).	Time-targeted notifications: push offers to customers in each zone during that zone's peak period (e.g., Ramanagaram and Govindraj Nagar at night, Hebbal 16-20).	Offer redemption, trips per notification
6	99 customers made 983 trips (about 10 each) and 30 drivers about 33 each.	Customer loyalty programme; driver tiers by rides and rating; consider growing the customer base, which looks small.	Active customers, rides per customer

Conclusion

The data tells one story: Namma Yatri does not have a "when" problem, it has a "conversion" problem. Demand and revenue are spread evenly across the day (only a 16% gap between busiest and quietest period) and payment behaviour is balanced. But only 45.5% of searches become rides, 23% of allotted rides cancel, and zone conversion ranges from 32% to 58%. Fixing the 8 weakest zones, tightening driver acceptance, and focusing marketing on high-fare, low-volume zones are the actions with the clearest payoff.

Limitations: the dataset is small (2,161 searches, 983 trips, 30 drivers, 99 customers) with no dates or timestamps on searches; fares are not related to distance, so revenue conclusions should be re-checked on a larger, validated dataset.